

Breaking through Accessed Bit Limits of DAMON

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LINUX

PLUMBERS CONFERENCE

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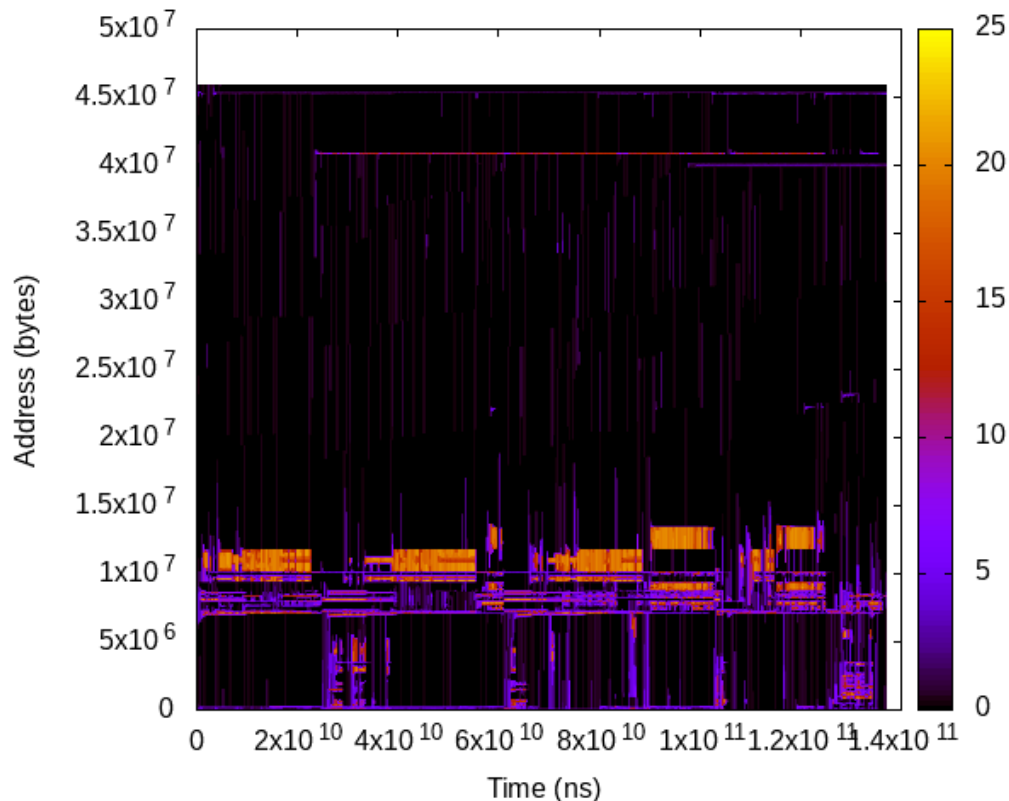


DAMON in One Slide

- Spawn a kernel thread that does
- See if each page is accessed, every 5ms
- Inform users the findings, every 100ms
- Time intervals can be [auto-]tuned

Providing Access Information

- Location
- Frequency
- Stability
- Recency



Access pattern heatmap of
Splash2x/Raytrace

Page Table Accessed Bit (A-bit)

- DAMON's access check primitive
- H/W sets the bit when it access the page
- S/W (DAMON) can clear the bit

A-bit Limits

- Lack of info
 - No accessor (which CPU/proc/cgroup)
 - No access type (read/write)
- Page table (reverse) walk overhead
- (Huge) Page granularity

Idea to Break-through Limits

- Extend DAMON for Data Attributes
- Attribute Examples
 - Memory is backed by cgroup X
 - A-bit of the data is set
 - AMD IBS found access to the data
(with accessor and access type info)

Roadmap

- Extend DAMON for Data Attributes
 - Milestone 1, by LPC'26
- Support perf event based access attribute
 - Milestone 2, by LSFMMBPF'27
- Support AMD IBS, Intel PEBS, ARM SPE, ...

Milestone 1: Complete

- DAMON is (almost) renamed to DAMON
 - Data Access/Attribute Monitoring and Operations eNgin
- DAMOS update is in progress

Milestone 2: In Progress

- Perf cycle event based access monitoring
 - Akinobu suggested initial form
 - Ravi suggested cross-vendor perf substrate
 - SJ is revisioning
- Decode perf sample load/store instruction

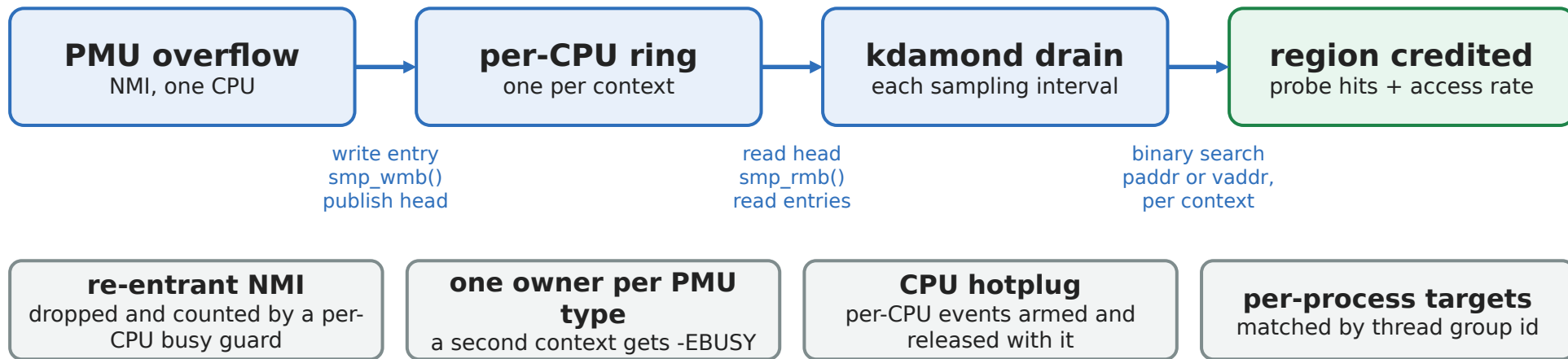
Milestone 3: Prototypes Ready

- Ravi: AMD IBS, Intel PEBS Support
- Kunwu: Arm SPE Support

Milestones 2&3, More Details

Milestone 2: NMI-Safe Delivery

PMU overflow runs in NMI; `damon_report_access()` takes a mutex, so delivery is deferred



Next, with Akinobu: fold into milestone 2's report path · `damon_report_access()` made lock-free

Milestone 3: IBS, PEBS, SPE

TODAY · ACCESSED-BIT SCAN

page-table walk
reads and clears the Accessed bit

one look per sampling interval
accessed or not, no more

THIS WORK · PERF EVENTS

Intel PEBS
virtual address

AMD IBS
physical address

Arm SPE
Kunwu Chan's backend

perf overflow handler · NMI context
turns each sample into an access report

AUX trace buffer
kdamond parses it, no NMI

per-context, per-CPU ring
lockless, NMI-safe

kdamond drain
each sampling interval

region access rate + per-probe hits

damon_hot_score() → DAMOS: promote · demote

Probe weight 0: alongside the scan, both raise the access rate. Probe weights set: scan off, weighted hits decide.

Built on [Akinobu Mita's perf-event access check](#) and co-developed with him · [RFC v3, posted October 2026](#)

Discussion Time

- Feel free to use
 - Hallway, eventing events
 - damon@lists.linux.dev
 - DAMON beer/coffee/tea [chat](#)
 - sj@kernel.org